Object Oriented Programming

Lecture 9

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Composition and Aggregation

Lets Recall

OOP: “its all about code reuse”

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<u>One way is to</u>

Use object of **one class** in **another class**

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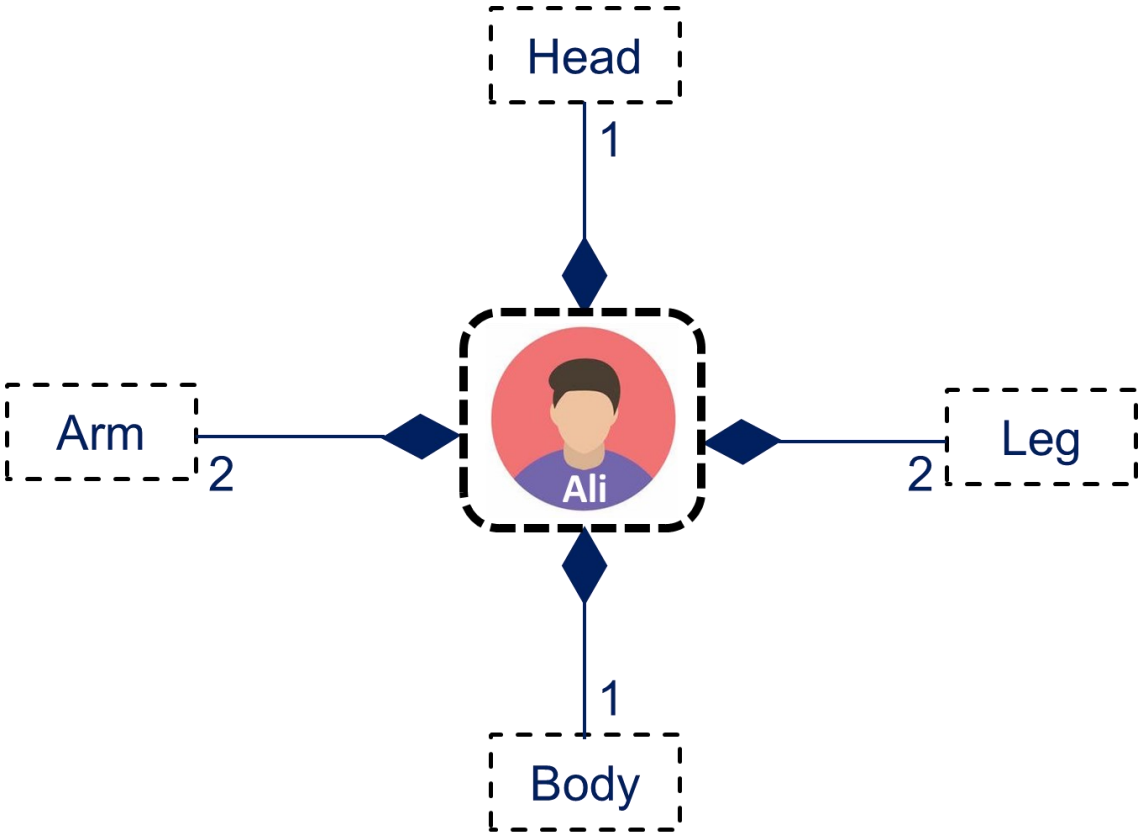
Composition

Aggregation

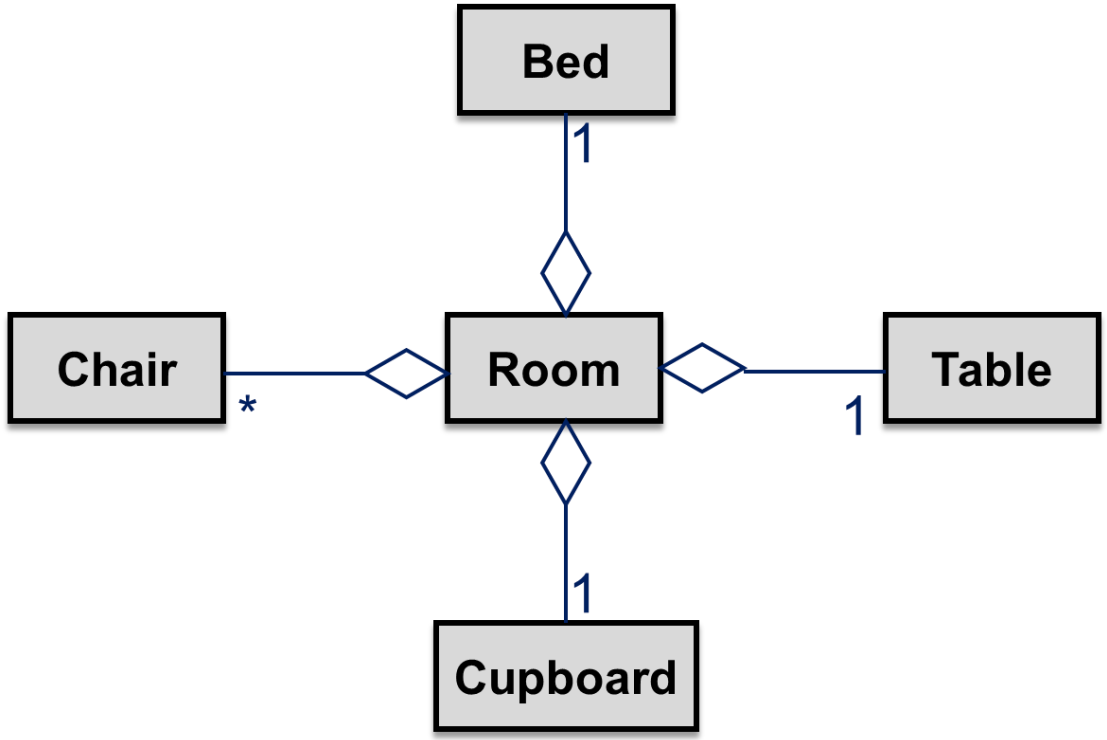
Lets Recall

OOP: “its all about code reuse”

Composition



Aggregation



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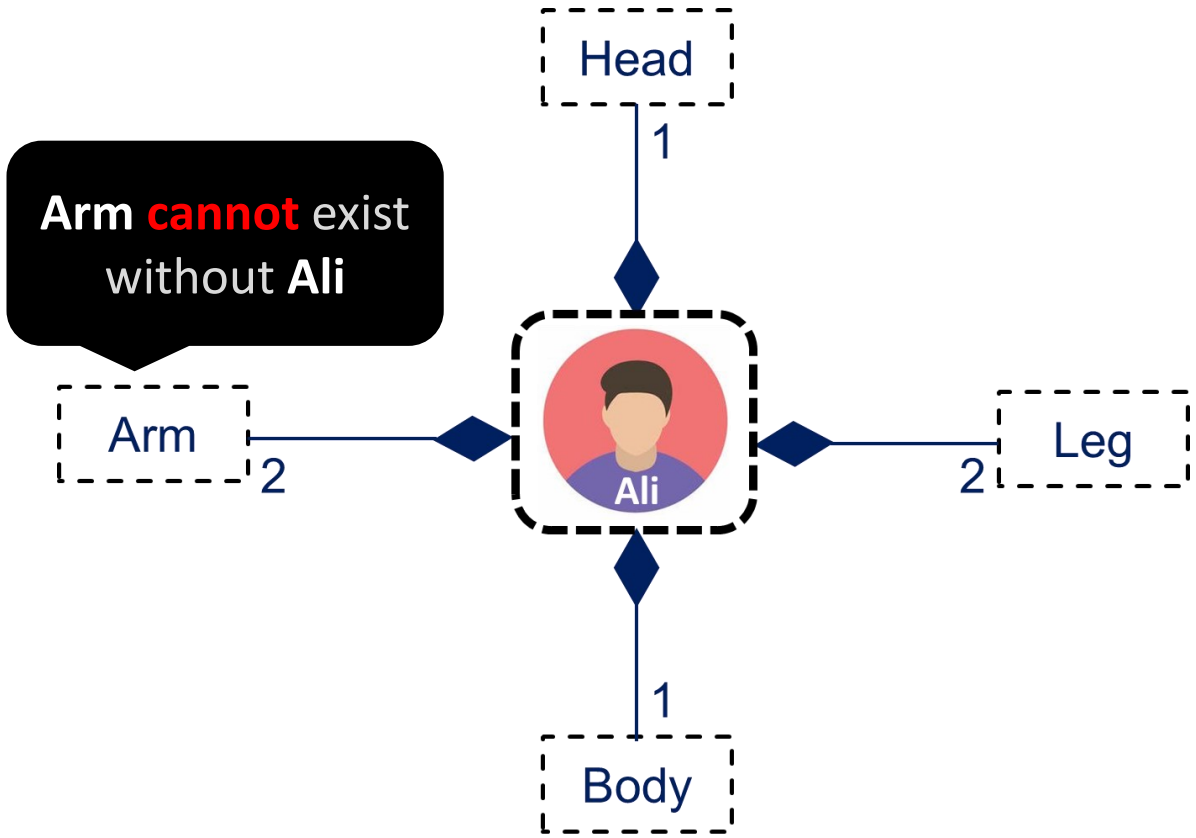
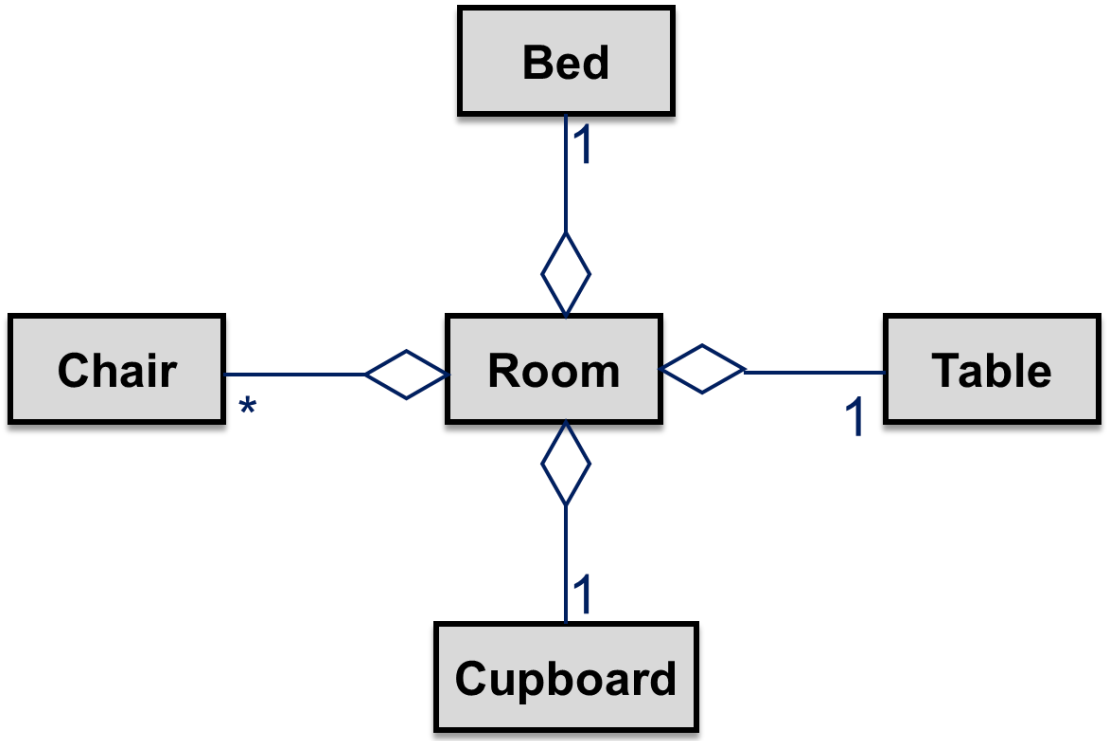


Aggregation



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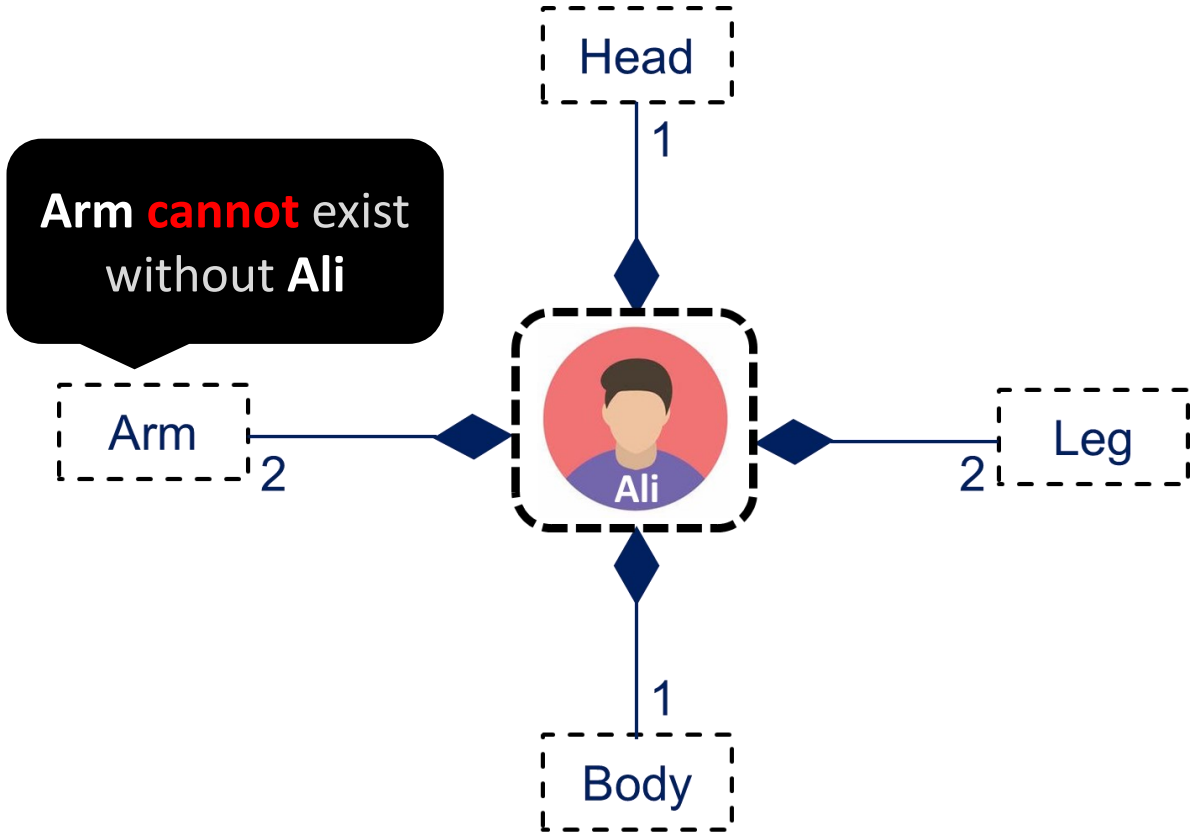

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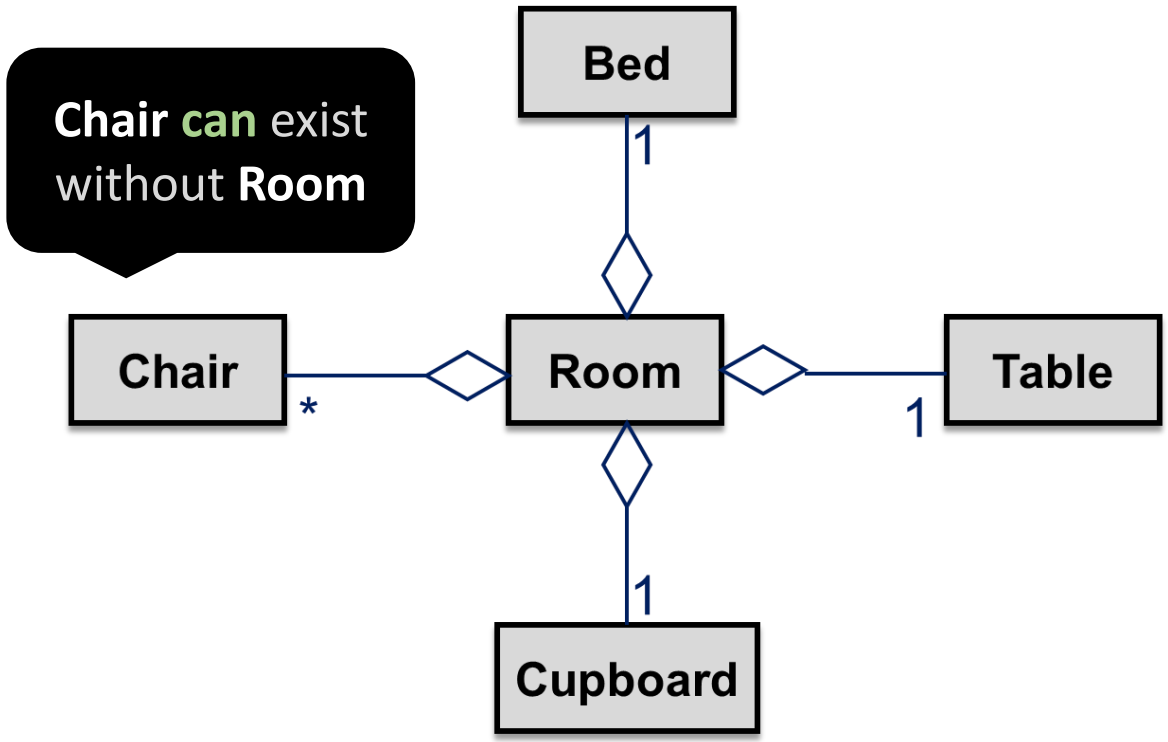

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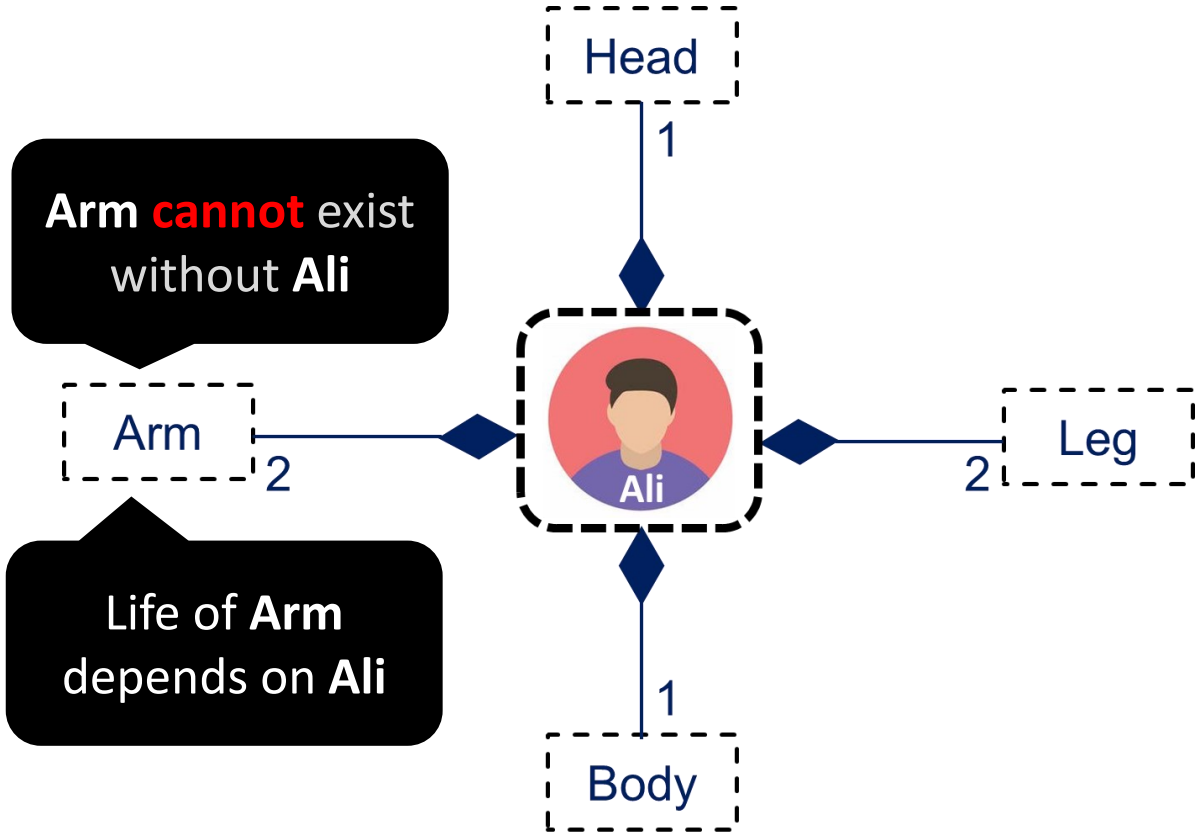

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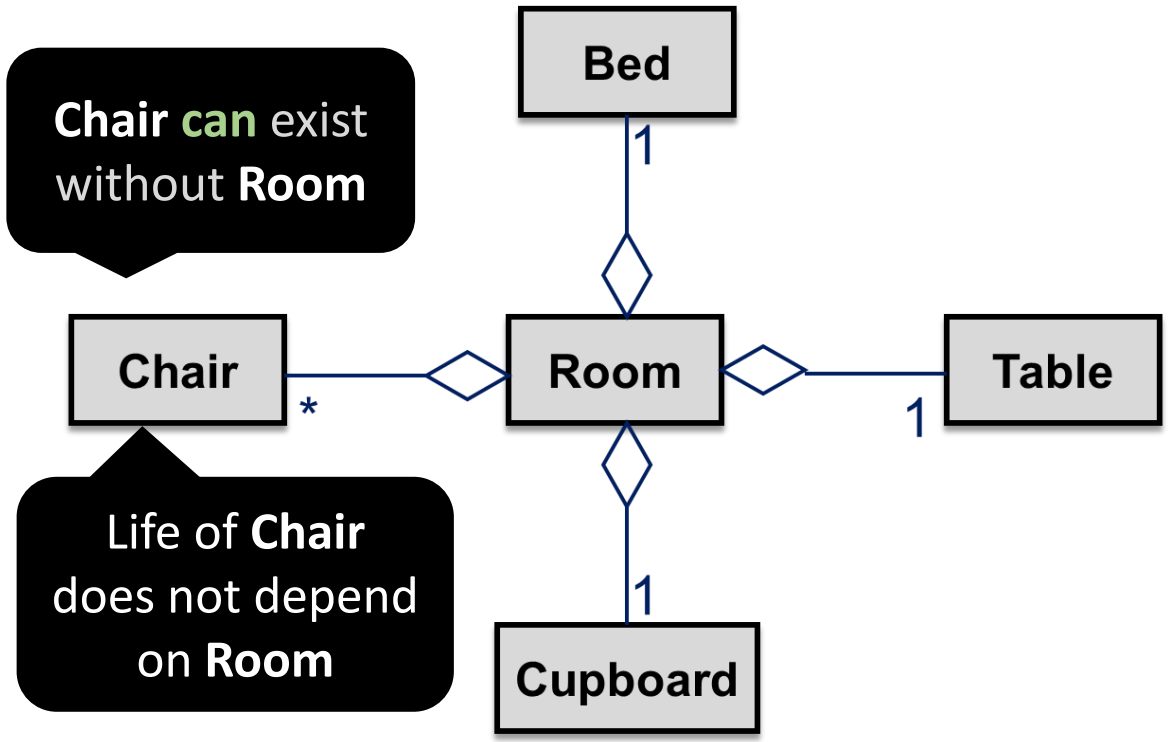

Lets Recall

OOP: “its all about code reuse”

Composition

Arm **cannot** exist without **Ali**

Head

1

Arm

2

Leg

2

1

Body

Life of **Arm** depends on **Ali**

Aggregation

Chair can exist without **Room**

Bed

1

Chair

Room

Table

1

1

Cupboard

Life of **Chair** does not depend on **Room**

What does it mean in C++ program

In C++

Composition

```
class Bar {
};

class Foo {
Private:
    Bar bar1;
}

void main {

    Foo foo1;
}
```

Aggregation

In C++

Composition

```
class Bar {
};

class Foo {
Private:
    Bar bar1;
}

void main {

    Foo foo1;
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```

Foo own **Bar** object and responsible for **Bar** lifetime. When **Foo** dies, so does **Bar**

Aggregation

In C++

Composition

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class Bar {
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    Foo foo1;
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```

Foo own **Bar** object and responsible for **Bar** lifetime. When **Foo** dies, so does **Bar**

Aggregation

```
class Bar {
};

class Foo {
Private:
    Bar* bar1;
    Foo(*Bar X)
    {
        bar1=X;
    }
}

void main {
    Bar a
    Foo foo1(&a);
}
```

In C++

Composition

```
class Bar {
};

class Foo {
Private:
    Bar bar1;
}

void main {

    Foo foo1;
}
```

Foo own **Bar** object and responsible for **Bar** lifetime. When **Foo** dies, so does **Bar**

Aggregation

```
class Bar {
};
class Foo {
Private:
    Bar* bar1;
Public:
    Foo(*Bar X)
    {
        bar1=X;
    }
}

void main {
    Bar a
    Foo foo1(&a);
}
```

Foo has an object which it borrowed from someone else. When **Foo** dies, **Bar** may live on.

In C++

Composition

```
class Bar {
};

class Foo {
Private:
    Bar bar1;
}

void main {

    Foo foo1;
}
```

Constructors of the sub-objects are always executed before the constructors of the master class

Foo own **Bar** object and responsible for **Bar** lifetime. When **Foo** dies, so does **Bar**

Aggregation

```
class Bar {
};
class Foo {
Private:
    Bar* bar1;
Public:
    Foo(*Bar X)
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}

void main {
    Bar a
    Foo foo1(&a);
}
```

Foo has an object which it borrowed from someone else. When **Foo** dies, **Bar** may live on.

In C++

Composition

```
class bar
{
public:
    bar()
    {
        cout << "I'm in bar" << endl;
    }
};

class foo
{
    bar b;
public:
    foo()
    {
        cout << "I'm in foo" << endl;
    }
};
```

```
int main()
{

    foo f;
}
```

Microsoft Visual Studio Debug Console

```
I'm in bar
I'm in foo
```

Thanks a lot

If you are taking a Nap, **wake up**.....Lecture Over